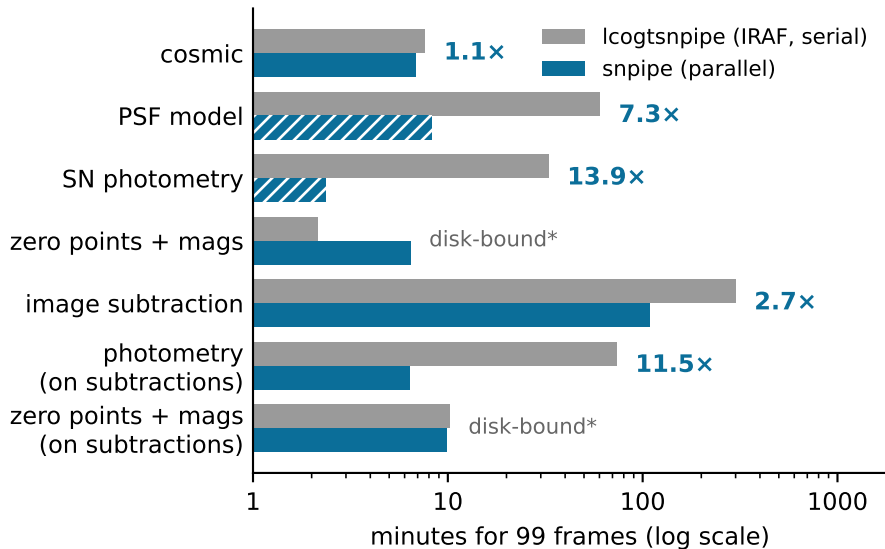
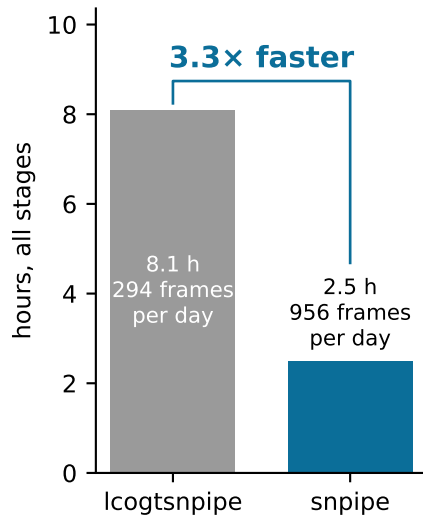


(a) per stage



(b) end to end



Same 99 LCO frames of SN 2024pxl, same 8-core / 16 GB node. Hatched: new time estimated from per-frame times. *Both pipelines read every full image once for the limiting magnitude; this step is limited by the shared disk and its time depends on disk load during the run, not on the code.